Anshu Nigam

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Career Objective

3rd-year B.Tech Computer Science (IoT) student seeking Software Engineering / AI-ML internship opportunities, bringing hands-on full-stack development, applied machine learning, and freelance client experience across production-style, deployed projects.

Education

Institute of Engineering and Management, Kolkata **2024 – 2028**
B.Tech in Computer Science and Engineering, specialization in IoT (CGPA: 8.36) *Kolkata, India*

Relevant Coursework

- Data Structures & Algorithms
- Object-Oriented Programming
- Computer Organization & Architecture
- Database Management Systems
- Statistics and Probability, Linear Algebra
- Artificial Intelligence, Machine Learning
- Deep Learning
- Operating Systems

Publications

HARD-RL: Reinforcement Learning for Fog Computing Load Balancing | *IJACSA*2026

- Developed a reinforcement learning framework for fog computing load balancing by extending Q-learning with a Fuzzy Logic Gate, adaptive reward mechanism, and energy-aware ϵ-decay.
- Improved task allocation reliability, achieving a **98.29% reduction in failed task allocations**.
- Reduced energy consumption by **32.91% per task** through energy-aware load balancing.
- **Publication Status:** Manuscript submitted to IJACSA; publication pending, 2026.

Projects

Full-Stack & Software Engineering

AutoBench — No-Code AutoML Platform | *FastAPI, Next.js 14, TypeScript* **2026**
∗ Architected a full-stack, no-code AutoML SaaS platform enabling users to train and benchmark models across 12 ML algorithms without writing code.
∗ Built Models, Datasets, and Experiments dashboards with real-time KPI stat cards, skeleton loading states, and podium-style leaderboards.

MeetFlow — Real-Time Video Conferencing App | *React.js, Node.js, WebRTC, Socket.io* **2026**
∗ Developed a real-time, peer-to-peer video conferencing app supporting multi-user rooms via WebRTC, avoiding the cost of a media-relay server.
∗ Engineered Socket.io signaling for real-time, room-based communication and connection handling across simultaneous participants.

Motion Amp — Motion Amplification Video Analysis (SIH Project) | *React (Vite), FastAPI* **2026**
∗ Building the frontend for a motion-amplification video analysis system as part of an SIH (Smart India Hackathon) team project, integrating with an existing FastAPI backend.
∗ Rebuilt the frontend on a Vite + React architecture for faster builds and a more modern development workflow.

Interactive Portfolio Site | *Next.js, Three.js / React Three Fiber, Framer Motion* **2026**
∗ Designed and built a gaming-controller-themed portfolio featuring a terminal-boot hero animation, an animated node-network canvas, and WebGL-based interactions.

Machine Learning & Data Science

Credit Card Fraud Detection System | *Random Forest, SHAP, FastAPI* **2026**
∗ Engineered a fraud detection model (Random Forest) achieving 0.99 AUC on a highly imbalanced dataset, layering SHAP-based explainability for interpretable predictions.

AI-Based Resume Parser & Candidate Ranking System | *Python, NLTK, Scikit-learn* **2026**
∗ Built an NLP pipeline (tokenization, stemming, lemmatization, TF-IDF + cosine similarity) layered with K-Means, KNN, and Decision Tree classifiers to rank candidates into a 0–10 job-fit score.

Handwritten Digit Recognizer | *Convolutional Neural Network (CNN)* **2026**
* Trained a CNN achieving 98.93% test accuracy on handwritten digit classification and deployed a live interactive demo on Hugging Face Spaces.

Telecom Customer Churn Prediction | *Logistic Regression, Random Forest, XGBoost, SMOTE* **2026**
* Benchmarked Logistic Regression, Random Forest, and XGBoost churn models, applying SMOTE to correct class imbalance and improve recall on the minority churn class.

Email Spam Detection | *SVM, TF-IDF, NLP* **2026**
* Developed an email spam classifier using SVM with TF-IDF vectorization and NLP preprocessing; documented in a Jupyter notebook and formal report.

Certifications

Apna College – Sigma 8.0 Batch (Completed) | CodeWithHarry – Data Science Course (Completed)

Achievements

- Secured a Top 10 finish at a college-level hackathon; competing in Smart India Hackathon (SIH) 2026 with the Motion Amp project.
- Self-directed ML curriculum implementing ensemble methods from scratch (Naive Bayes through Gradient Boosting, XGBoost, and Stacking), structured around an 8-phase AI/ML roadmap from math foundations through generative AI/LLMs.
- Solving Data Structures & Algorithms problems in Java, building toward placement and technical-interview readiness.
- Maintains a public GitHub portfolio spanning full-stack engineering, NLP, computer vision, and reinforcement learning projects.